

Multiple Choice (10 marks)

1. By definition, the electric displacement current through a surface S is proportional to the
 - (a) magnetic flux through S
 - (b) rate of change of the magnetic flux through S
 - (c) time integral of the magnetic flux through S
 - (d) rate of change of the electric flux through S
2. A refracting telescope consists of two converging lenses separated by 100 cm. The eyepiece lens has a focal length of 20 cm. The angular magnification of the telescope is
 - (a) 4
 - (b) 5
 - (c) 6
 - (d) 20
3. Which of the following is true about any system that undergoes a reversible thermodynamic process?
 - (a) There are no changes in the internal energy of the system.
 - (b) The temperature of the system remains constant during the process.
 - (c) The entropy of the system and its environment remains unchanged.
 - (d) The entropy of the system and its environment must increase.
4. Consider an electron in the $n = 4$, $l = 1$ state in hydrogen. Which of the following final states can NOT be reached by an allowed transition?
 - (a) $n = 3$, $l = 2$
 - (b) $n = 3$, $l = 1$
 - (c) $n = 3$, $l = 0$
 - (d) $n = 2$, $l = 0$
5. A single-electron atom has the electron in the $l = 2$ state. The number of allowed values of the quantum number m_l is
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 5

True / False (10 marks)

Answer True or False

6. True or False: The resolving power of the grating spectrometer for wavelengths of 500 nm and 502 nm is 250.
7. True or False: In a single-electron atom with $l = 2$, there are 5 allowed values for the magnetic quantum number m_l : -2, -1, 0, 1, and 2.
8. True or False: Cooper pairs in superconductors are primarily attracted to each other through the strong nuclear force.
9. True or False: The Hall coefficient of a doped semiconductor indicates the sign of the charge carriers present in the material.
10. True or False: If the absolute temperature of a blackbody is increased by a factor of 3, the energy radiated per second per unit area increases by a factor of 81.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process and calculations involved in determining the final voltage across a charged capacitor when it is connected to two uncharged capacitors in series.
12. Discuss the principles of length contraction in special relativity. Calculate the required velocity for an observer to measure a 1.00 m rod as 0.80 m when moving parallel to it.

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